

PDF_263 — Yang–Mills Tower Equation Report

An Honest, Complete Audit of the YM/OS Witness-Collapse Bricks

Theorema Aureum 143 (Volume I) — Morning Star Project

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HONESTY NOTICE (read first). This report makes *no* claim that the Yang–Mills mass gap is proven, that $\mu > 0$, or that Clay “Surface #1” is closed. Every one of the 13 modules audited here is currently an **OPEN** (named, *unproven*) proposition. The axiom footprint is the classical trio {`propext`, `Classical.choice`, `Quot.sound`} — *not* the empty set. There are no `sorry/sorryAx/admit proof-terms`. This is logical-hygiene bookkeeping; it proves nothing new. Sections that earlier drafts requested but for which *no artifact exists on disk* (an “LLM error / Z-protocol” narrative and “Besel I” test logs) are documented as **absent** in §4 rather than fabricated.

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1 Equation statement — the Yang–Mills mass-gap surface

The Clay “Yang–Mills existence and mass gap” problem asks, for a compact simple gauge group (here $SU(3)$), for a quantum Yang–Mills theory on $\mathbb{R}^4$ whose Hamiltonian has a strictly positive spectral gap above the vacuum. The chain of statements the YM/OS modules in this repository attempt to encode is the following.

(E1) Exponential clustering (two-point decay).

$$\left| \langle W(C_1) W(C_2) \rangle - \langle W(C_1) \rangle \langle W(C_2) \rangle \right| \leq C e^{-\Delta \text{dist}(C_1, C_2)}, \quad \Delta > 0.$$

- $W(C)$ — the Wilson loop observable on a closed contour C (the gauge-invariant trace of the parallel transport around C).
- $\langle \cdot \rangle$ — the (Euclidean) expectation in the lattice Yang–Mills measure.
- $\text{dist}(C_1, C_2)$ — the separation between the two loops.
- $\Delta > 0$ — the *mass gap*: the rate of exponential decay of connected correlations. Strict positivity of Δ is the entire content of the problem.
- C — a finite positive constant.

(E2) Transfer-operator gap. Clustering at rate Δ is equivalent to a spectral gap of the Euclidean transfer operator T_{real} acting on the physical Hilbert space:

$$\text{spec}(T_{\text{real}}|_{\mathcal{H}_0}) \subseteq [0, \lambda_1], \quad \lambda_1 = \|T_{\text{real}}|_{\mathcal{H}_0}\| < 1, \quad \Delta = -\log \lambda_1 > 0.$$

- T_{real} — the genuine Wilson/heat-kernel transfer operator (one time-step Euclidean evolution).
- $\mathcal{H}_0$ — the orthogonal complement of the vacuum (mean-zero sector).
- λ_1 — the operator norm of T_{real} restricted to $\mathcal{H}_0$; the second-largest eigenvalue modulus.
- $\Delta = -\log \lambda_1$ — the same mass gap as in (E1).

(E3) Mass gap from the gap bound. A self-adjoint T on a Hilbert space H has a mass gap m when

$$\langle x, Tx \rangle_{\text{re}} \geq (1 - m) \|x\|^2 \quad \text{fails strictly on } \mathcal{H}_0, \quad \text{equivalently } \lambda_1 \leq e^{-m}, \quad m > 0.$$

This is the predicate the repository calls `HasMassGap H T m`.

The honest gap. A *non-vacuous* discharge of (E1)–(E3) requires a real $SU(3)$ Wilson transfer operator with a strictly positive lower bound $T_{\text{real}} > 0$. The ingredients — $SU(3)$ character theory / a verified heat-kernel transfer operator — are **absent from mathlib v4.12.0**. Consequently **no $T_{\text{real}} > 0$ is constructible in this repository**, and every module below was discharged only at a *degenerate witness* (a *witness collapse*): the correlator taken $\equiv 0$, the operator $T := 0$ on $H := \mathbb{C}$, or a reflexive definitional re-wrap. The recorded T_{real} lower bound for all 13 modules is therefore **none**.

2 Step-by-step audit — every YM/OS witness-collapse brick

The table below lists all 13 modules audited on the **2026-06-03** repair pass. Each was previously registered as a “brick” (a landed result in `scripts/check-towers.sh`); each has now been converted *in place* from a proved theorem/lemma into a named open proposition `def <name>_OPEN : Prop := <full proof>` and de-registered from the BRICKS array. Status for all: **OPEN** (witness-collapse), T_{real} bound = **none**.

#	File	Named open Prop	Witness collapse
1	YM/ClusteringCore.lean	clustering_zero _OPEN	clustering predicate instantiated at constant-zero pair (vacuous)
2	YM/MassGapStandin.lean	mass_gap_standin_example _OPEN	mass-gap witness is the zero CLM on $H := C$ (vacuous)
3	YM/SpectralGapCore.lean	hasMassGap_zero _OPEN	HasMassGap at $T := 0$ collapses to $0 \leq (1-m)*\text{norm}^2$ (vacuous); def HasMassGap preserved
4	YM/TransferOperator.lean	transfer_gap _OPEN	transferGapBound at degenerate operators (vacuous); def transferGapBound preserved
5	YM/TwoPointDecay.lean	clustering_zero_from_transfer _OPEN	clustering of constant-zero function, hypothesis unused (vacuous); def hasExponentialClustering preserved
6	YM/MassGapFromDecay.lean	mass_gap_from_clustering_zero _OPEN	mass gap from constant-zero clustering, zero witness operator (vacuous)
7	YM/IntegratedTailReal.lean	integrated_tail_le_exp _OPEN	integrated_tail $L m := \text{rexp}(-m*L)$ so bound is reflexive $\text{rexp}(-m*L) \leq \text{rexp}(-m*L)$ (vacuous); def integrated_tail preserved
8	YM/TransferGapReal.lean	transfer_gap_real _OPEN	real-named transfer gap still discharged at degenerate witness (vacuous)
9	YM/MassGapReal.lean	mass_gap_from_transfer _OPEN	mass gap from transfer at zero witness on $H := C$ (vacuous)
10	YM/ClusteringImpliesGap.lean	clustering_implies_gap _OPEN	constant-zero antecedent, zero CLM consequent (vacuous)
11	YM/TransferImpliesClustering.lean	transfer_implies_clustering _OPEN	constant-zero witness is constant-zero function, hypothesis unused (vacuous)
12	YM/TailImpliesTransfer.lean	tail_implies_transfer _OPEN	integrated_tail and transfer-GapBound unfold to same inequality; pure definitional re-wrap (vacuous)
13	YM/GapToDecay.lean	gap_to_decay _OPEN	consequent discharged by $f t := \text{exp}(-m*t)$, gap hypothesis unused (vacuous)

Reading the columns. WHAT — the file path and the named proposition (namespace `TheoremaAureum.Towers.YM.OS`). **WHY** — the proposition’s role: each is one link in the $(E1) \rightarrow (E2) \rightarrow (E3)$ clustering/transfer/mass-gap chain of §1. **WHERE** — the $T_{\text{real}} > 0$ status: **none** for all, because the proof discharges only at a degenerate witness. **WHEN** — the 2026-06-03 repair audit. The full per-file SHA-256 hashes are in Appendix 6.

3 Representative Lean excerpts (verbatim) with English gloss

These are exact, unedited excerpts of the converted source on disk. They illustrate the three collapse shapes; the remaining ten modules collapse for the analogous reasons recorded in §2.

3.1 Mass-gap inhabitedness (MassGapStandin.lean)

```
def hasMassGapLowerBound (mu : R) : Prop :=
  exists C : R, 0 < C and 0 < mu

def massGap_standin_example_OPEN : Prop :=
  hasMassGapLowerBound 1
```

Gloss. The predicate asserts only “there exist a positive constant C and a positive mass μ .” Witnessing it at $C = \mu = 1$ says nothing about any Yang–Mills correlator. As written the `_OPEN` proposition is trivially satisfiable. (ASCII rendering: $R = \mathbb{R}$, `exists` = $\exists$, `and` = $\wedge$, `mu` = μ .)

3.2 Reflexive tail bound (IntegratedTailReal.lean)

```
noncomputable def integrated_tail (L m : R) : R := rexp (-m * L)

def integrated_tail_le_exp_OPEN : Prop :=
  forall (L m : R), 0 <= m -> 0 <= L -> integrated_tail L m <= rexp (-m * L)
```

Gloss. Because `integrated_tail L m` is *defined* as e^{-mL} , the “bound” unfolds to the reflexive $e^{-mL} \leq e^{-mL}$ and closes by `le_refl`. The symbol does not refer to any heat-trace integral; the genuine statement (a real heat-trace tail bounded by an exponential) is unreachable here.

3.3 Gap $\Rightarrow$ decay (GapToDecay.lean)

```
def gap_to_decay_OPEN.{u} : Prop :=
  forall {H : Type u} [NormedAddCommGroup H] [InnerProductSpace C H] (m : R),
    0 < m -> m < 1 ->
    (exists T : H ->L[C] H, T != 0 and HasMassGap H T m) ->
    hasExponentialClustering (fun t => Real.exp (-m * t)) m
```

Gloss. The consequent is discharged by the obvious function $f(t) = e^{-mt}$ (which trivially satisfies $|f(t)| \leq 1 \cdot e^{-mt}$) with the spectral-gap hypothesis *unused*. So the implication holds as written without using its antecedent — a witness collapse. It does not show a real correlator decays from a real spectral gap.

4 Requested sections with no artifact on disk (documented, not fabricated)

Earlier drafts of this report requested an “LLM Error Information: Z is a Protocol” section and a “Besel I concretization” section, each to cite test runs and a data file. The honesty rule of this project forbids presenting unrecorded narrative or test results as data. The following were checked on disk on 2026-06-03 and found **absent**:

- `symmetry_error_rate.csv` — **not present** anywhere in the workspace.
- `.local/Besel_I/` (or any `*esel*` directory) — **does not exist**.
- `/mnt/data/Report_263.pdf` / `/mnt/data/PDF_263_*` — `/mnt/data` is **not part of this workspace** (workspace root is `/home/runner/workspace`); no prior draft to merge exists.

There is no recorded “Z-protocol” study, “phase-shift event”, “Spark Muse” observation, attributed quotation, or “Besel I” test log in the repository. Because none of these artifacts exists, these sections are **omitted**. Stating their absence here is the honest treatment: the report leaves out no *true* implication, and fabricates none.

5 Compliance certificate — honest

Property	Verified value
Modules audited	13 (all YM/OS)
Non-vacuous $T_{\text{real}} > 0$ proofs achieved	0
Converted to named open Prop	13
De-registered from BRICKS	13
Axiom footprint (each module)	classical trio: <code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>sorry</code> / <code>sorryAx</code> / <code>admit proof-terms</code>	0 / 0 / 0
Compile (direct-lean)	all EXIT 0
Surface #1 (global YM mass gap)	OPEN

Honest caveats on this certificate.

1. **The axiom footprint is the classical trio, not the empty set.** Each `_OPEN` proposition references `mathlib (Measure/inner/Real.exp)`, which pulls in `{propext, Classical.choice, Quot.sound}`. “No axiom” would be *false*; “no *research-grade* axiom beyond the trio” is the true statement.
2. **A literal grep is *not* empty.** The strings `sorry`, `admit`, and `axiom` occur in the CLAY OPEN docstrings of these files (and across the wider tree — 140 files contain at least one of those strings). The honest signal is therefore the *proof-term* count (0) plus `#print axioms` (trio only), **not** an empty grep.
3. **lake build was not run.** The vendored `mathlib v4.12.0` tag is unresolved, so any `lake/lake env` invocation would re-fetch from remote and wipe the oleans. Verification used the documented direct-lean bypass (Appendix 6). No “lake build exits 0” claim is made.
4. **The `_OPEN` propositions are trivially satisfiable as written.** The degenerate witnesses remain available, so each `_OPEN Prop` *names* the as-written shape but does **not** assert the genuine YM surface. The conversion removes a vacuous landed *theorem*; it manufactures nothing.

Status of the other towers (no per-brick claim made here). This report audits the 13 YM/OS modules only. For the record, and per the project’s locked invariants: the YM and Navier–Stokes towers remain **Status: Open** at their Clay surfaces (YM Surface #1 open; NS Surfaces #1/#2 open, NS frozen); the RH fragment in `Towers/RH/` is a *conditional reduction* over two undischarged open hypotheses, not a proof of the Riemann Hypothesis. No tower in this repository closes a Clay problem.

6 Appendix — provenance and verification method

Verification method (verbatim from the build manifest).

- Method: direct-lean (LEAN_PATH over existing oleans; NOT lake).
- Reason: vendored mathlib v4.12.0 tag unresolved; lake env would re-fetch and wipe oleans.
- Axiom check: #print axioms appended to /tmp copy, compiled with lean ;file;.
- All EXIT 0: True; all trio-only: True.

Per-file SHA-256 (full).

#	File (under Towers/)	SHA-256
1	YM/ClusteringCore.lean	f3ceecd3892edd81a760aa8205e947aef57ee273c84ef5be33c3300f8aa9a1fb
2	YM/MassGapStandin.lean	ffa6f5820e6d6fdc6b0af4051582be2508a35e21117d87e24f26eb799e8ee5d
3	YM/SpectralGapCore.lean	fca1de2b085e768a61ef320db8d7d23f546a0f3ddce0b76ccbc3f956eee69e89
4	YM/TransferOperatorBound.lean	aff0bb4887885a10e11ef0e41a46dd801acb0ff5cb58b75d3d613a3d27e10295c
5	YM/TwoPointDecay.lean	516b8fb721a762181a0200f11e8fd01f16ee01c19ae8228fca944ab907234575
6	YM/MassGapFromDecay.lean	b5acbbc0770f6211765989440ec82a88d6d2744ae7a5d45d383eaa1360adaa3b
7	YM/IntegratedTailReal.lean	223a0dacab5f7bc4b871b9a1dbe620d7944af091d7bbd65d6f6f693d68ef8ca2
8	YM/TransferGapReal.lean	bd30cd0f17d3a74ae8870c1fc2808a9584d6e77185274760367abda5484ebe
9	YM/MassGapReal.lean	30f0b1b37c510cee3265a1f5e3e2bccc04647a500273f28a09902ffcf51fb0
10	YM/ClusteringImpliesGap.lean	acc573093f6dc6b68e7496ebfd99932ba91a63937a5430c5f4e5eadea3ba1626
11	YM/TransferImpliesClustering3a502b	acd4110ed8e52b5ad1841e98c0e45164e974756b013957b092bff7a301
12	YM/TailImpliesTransfer.lean	e0d3ba5d3cbcad9f17cea42a73cc155af00af2dbb91fc0254f70c8016643bd50
13	YM/GapToDecay.lean	ff4b8f5dd357a353982afe08e66afd8036d4eabc526af4d0c8c364828f235dbe

Source documents. BUILD_MANIFEST.v2.7.json, CLAY_STATUS.md, reports/CLAY_REPAIR_2026-06-03.md, provenance/clay_repair.diff. This PDF was generated by reports/build_pdf263.py directly from those files; every numeric value above is copied from them or from a command whose result is stated in the body.